# Automated Credit Scoring, Fairness, and Ethical Decision Design

*To what extent can an automated credit scoring model improve default prediction while remaining ethically defensible, once we account for fairness trade-offs, calibration, and how predicted probabilities are converted into real decisions?*

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## 1. Introduction

Access to credit shapes life opportunities, influencing housing choices, education and so on. For this reason, credit scoring has long sat at the boundary between statistical prediction and social policy (Thomas et al., 2002; Anderson, 2007). In recent years, lenders have increasingly relied on data-driven systems that improve efficiency and ranking accuracy, yet they also introduce opacity and new forms of discrimination, particularly when predictive features correlate with protected characteristics or structural disadvantage (Berg et al., 2020; Bartlett et al., 2021).

This project asks whether gains in predictive performance remain ethically defensible once the full decision pipeline is taken seriously. Rather than treating any single fairness metric as authoritative, the analysis critically examines how model design, threshold choice, and calibration jointly determine who is denied credit, who is exposed to risky borrowing, and which groups ultimately bear the costs of error.

The starting point is that fairness cannot be reduced to a technical property of a model. While statistical criteria such as demographic parity, equal opportunity, and equalised odds provide useful auditing lenses, they embed contested value judgements about which disparities matter and which errors are considered most harmful (Hardt et al., 2016; Barocas et al., 2019). In addition, when group-based rates differ, the main fairness desiderata cannot all be satisfied at once, especially when probabilities are calibrated (Kleinberg et al., 2016; Chouldechova, 2017). As a result, fairness claims that do not explicitly justify their normative assumptions and operating point risk being incomplete or misleading.

## 2. Literature and conceptual foundations

Credit scoring research highlights two practical messages. First, even relatively simple models can perform well when they are built on strong predictors and are embedded in a clear decision process (Hand and Henley, 1997; Thomas et al., 2002). Second, performance summaries can mislead when the default is a rare event. This motivates evaluation tools that remain informative under imbalance and motivates threshold choice that is tied to costs rather than convention (King and Zeng, 2001; Elkan, 2001).

Algorithmic fairness research adds a complementary perspective. Group-based metrics are attractive because they are auditable and communicable, yet they can be satisfied in trivial ways and may conflict when groups face different underlying risk levels (approving or denying almost everyone). (Barocas et

al., 2019). These limitations motivate treating threshold selection as part of ethical design rather than as a technical afterthought.

A further limitation is that correlational audits alone cannot distinguish genuine differences in underlying risk from bias introduced by data generation or institutional practice. Causal approaches to fairness are often proposed as a stronger way to separate these mechanisms, but implementing them would require additional assumptions and design choices beyond the scope of this report. Accordingly, the results here should be read as descriptive evidence and ethical argument of disparity rather than as causal claims about discrimination or its sources (Barocas et al., 2019).

Finally, governance and transparency matter because ethical evaluation depends on how a model is documented and monitored. Datasheets and model cards encourage practitioners to state intended use, report subgroup behaviour, and record limitations as part of responsible deployment (Gebru et al., 2018; Mitchell et al., 2019). In the UK context, regulatory guidance on AI and data protection supports expectations of accountability, fairness, and meaningful information in automated decision-making (ICO, 2023).

## 3. Data and modelling approach

Credit scoring is not purely a forecasting task, because scores shape access to credit, predictive performance and ethical legitimacy must be evaluated jointly. The model studies a binary default risk classifier built on a reproducible 5,000-row random sample from the Home Credit Default Risk dataset. The sample was randomly partitioned into a training set (n = 4,000) and a held-out test set (n = 1,000) (Home Credit Group, 2018).

The outcome variable is TARGET, coded as 1 for default and 0 otherwise. The default rate in the 5,000-observation sample is 0.0846, reflecting the strong class imbalance typical of credit risk settings.

Two logistic regression models are estimated, as logistic regression is standard in credit scoring and offers transparent, auditable interpretation through log-odds and odds ratios, helping link modelling choices to credit risk practice (Hosmer et al., 2013; Anderson, 2007). The baseline model relies primarily on financial capacity and stability indicators, including age, income and loan measures, household composition, employment tenure, and administrative stability proxies. The extended model adds contextual and demographic features, regional rating variables, social circle indicators, and external

risk scores (EXT_SOURCE_1 to EXT_SOURCE_3), achieves substantially higher-ranking performance than the baseline (AUC 0.766 versus 0.655), but at the chosen threshold it identifies very few true defaults (default TPR 0.0108), resulting in near-universal approval (0.5).

These EXT_SOURCE variables can therefore behave like imported black boxes: the model not only learns from Home Credit application fields but also inherits judgments embedded in upstream scoring pipelines that are not in the dataset. If these external scores were constructed using indirect signals such as digital traces or geographic proxies, the model could be outsourcing bias while making the resulting decisions harder to explain or contest. Responsible modelling should therefore treat data provenance and explainability as integral to design rather than as an afterthought (Gebru et al., 2018; ICO, 2023).

Performance is assessed using receiver operating characteristic (ROC) curves and the area under the curve (AUC). AUC can be interpreted as the probability that a randomly chosen defaulter receives a higher predicted risk than a randomly chosen non-defaulter (Hanley and McNeil, 1982; Fawcett, 2006). To connect scoring to decisions, the report assessed outcomes at a fixed 0.50 threshold, with applicants above the threshold treated as default/denied.

Fairness is evaluated across gender, age bands, and regional rating groups using approval rates, disparate impact ratios, error rate gaps that relate to equal opportunity and equalised odds, and calibration overall and by group. Calibration is included because many credit systems use probabilities for pricing and limit setting, not only for accept-reject decisions, and miscalibration can therefore produce distributive harms even when approval rates look similar (Brier, 1950; Niculescu-Mizil and Caruana, 2005). This design supports an ethical comparison because the extended model's largest gains appear to come from the external scores, which are influential but not directly interpretable within the dataset.

Despite this, systematic differences emerge in score distributions, false positives, and calibration across groups. These findings highlight that fairness claims cannot be separated from threshold choice and downstream use. Responsible deployment should treat thresholding, calibration, and reporting as part of a socio-technical decision design, supported by transparent documentation and governance safeguards (Barocas et al., 2019; Mitchell et al., 2019).

## 4. Predictive performance and error profile

The extended model improves ranking performance relative to the baseline model, increasing AUC from 0.6552 to 0.7656. Figure 1 shows the ROC curves for both models (Baesens et al., 2003; Lessmann et al., 2015).

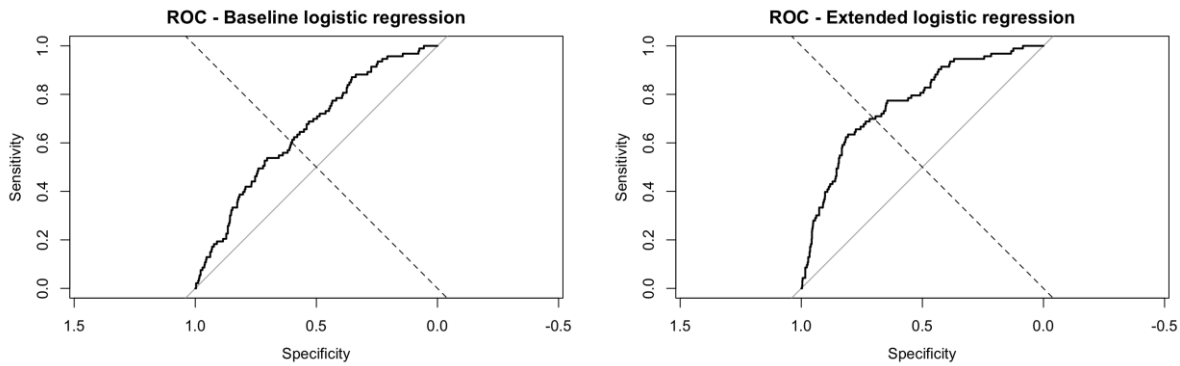



*Figure 1A. ROC curve for baseline logistic regression. Figure 1B. ROC curve for extended logistic regression.*

Table 1 summarises test set performance at the 0.50 threshold. The default positive class is TARGET = 1.

| Model | AUC | Accuracy (thr=0.50) | TPR for No Default[1] (thr=0.50) | TPR for Default[2] (thr=0.50) |
|---|---|---|---|---|
| Baseline | [0.6552] | [0.9070] | [1] | [0] |
| Extended | [0.7656] | [0.9040] | [0.9956] | [0.0108] |

*Table 1: Test-set performance metrics for the baseline and extended models.*

[1] We define pred_class with levels c("No Default", "Default") in the R code, yardstick::sens() reports TPR for No Default (equivalently specificity for Default).
[2] yardstick::spec() reports TPR for Default (equivalently sensitivity for Default).

Accuracy is therefore a fragile summary at this operating point because default is rare: with only about 8 per cent of applicants defaulting, a classifier that predicts non-default for everyone would already appear highly accurate. At the 0.50 threshold, the baseline model effectively behaves in this way, while the extended model differs only marginally. Both models classify almost all applicants as non-default: the baseline model predicts no defaults in the test set, and the extended model predicts only five. As a result, default detection is extremely weak, with a true positive rate of 0 for the baseline and 0.0108 for the extended model. This motivates reporting ranking measures such as AUC and cautions against interpreting high accuracy as evidence of effective default identification (King and Zeng, 2001). Rather than treating this as a technical bug, it is more revealing to view it as a conservative decision design that makes denial difficult unless the model assigns a very high default probability (Saito and Rehmsmeier, 2015).

This operating point has competing ethical interpretations. On one hand, denying very few applications reduces immediate exclusion risk, which matters when denial is treated as a primary harm. On the other hand, approving almost everyone weakens the protective role of screening and may increase the chance that vulnerable borrowers are approved for unaffordable debt. For that reason, the report treats thresholding as a policy lever that must be defended in normative terms, rather than being treated as a neutral technical default (Barocas et al., 2019; Thomas et al., 2002).

This matters because credit decisions can create harm in more than one direction. A system that is deployed at an operating point where it is largely insensitive to default risk may still look statistically competent in headline accuracy, yet it effectively transfers uncertainty from the lender to applicants. In ethical terms, this raises a duty of care concern: the institution is not merely allocating a scarce product, but also deciding whether to expose people to debt that they are unlikely to be able to repay.

At the same time, it would be equally problematic to swing to the opposite extreme and use a very low threshold to deny large numbers of applications. Widespread rejection can exclude households from mainstream credit, push some applicants towards more expensive or less regulated alternatives, and limit their ability to build a credit record. Conversely, near-universal approval can concentrate the longer-run costs of default on borrowers through damaged credit histories and reduced access to future credit, housing, and sometimes employment screening. For that reason, the ethical question is not whether approvals are high or low in isolation, but whether the decision rule balances access, protection, and accountability in a defensible way.

## 5. Fairness at the operating threshold

Tables 2A–2C report group-level outcomes for the extended model at the 0.50 threshold across gender, age bands, and regional rating. Approval rates are uniformly high across all subgroups, giving the appearance of strong demographic parity. However, this parity is largely uninformative, as it is driven by near-universal approval rather than by substantively justifiable decision-making. At this threshold, apparent parity reflects passive fairness, achieved mainly because the model does not act on the predicted differences (Barocas et al., 2019; Dwork et al., 2011). This pattern illustrates a well-documented ethical limitation highlighted in credit scoring: fairness metrics that depend on a fixed threshold can obscure underlying inequities rather than resolve them. From a consequentialist perspective, such a threshold may appear acceptable if it yields strong aggregate performance, even when harms are unevenly distributed across groups. Nevertheless, this assessment can obscure the fact that certain subpopulations may consistently bear disproportionate burdens, indicating that high overall performance does not guarantee equitable outcomes at the subgroup level. Moreover, reliance on a single operating point may conceal latent model biases that only become evident under alternative thresholds or in different contexts.

In contrast, from a deontological perspective, the same threshold is ethically problematic: once predictions are binarised for decision-making, it can generate unjust denials through opaque or indirectly discriminatory rules. Even when parity is achieved on one criterion, it may fail to generalise beyond a specific operating point and can reproduce disparate treatment once predictions are binarised for decision use (Casas et al., 2024). Collectively, these considerations illustrate that consequentialist and deontological frameworks provide complementary yet distinct ethical insights: the former emphasises aggregate outcomes and overall harm, whereas the latter underscores procedural justice and the protection of individual rights.

For gender, similar outcomes arise because few applicants are denied, reflecting equality of outcome rather than opportunity, even if score distributions differ (men and women). A stronger requirement would ask whether applicants with the same predicted risk are treated similarly, and whether error rates are comparable once denials occur. Evidence suggests this condition may fail that false positive and false negative imbalances can persist even after accounting for base rate differences, indicating that threshold-based parity can conceal ethically significant error asymmetries (Menghani et al., 2023). Under a very strict threshold, the model can become effectively blind to nuanced differences, so parity

claims should be interpreted with caution rather than taken as evidence of substantive fairness (Hardt et al., 2016).

*Table 2A. Outcomes by gender at threshold 0.50.*

| Group | n | Base default rate | Mean score | Approval rate | TPR | FPR | Accuracy |
|---|---|---|---|---|---|---|---|
| F | 625 | 0.0816 | 0.0729 | 0.997 | 0.0000 | 0.00348 | 0.915 |
| M | 375 | 0.1120 | 0.1050 | 0.992 | 0.0238 | 0.00601 | 0.885 |

*Table 2B. Outcomes by age band at threshold 0.50.*

| Group | n | Base default rate | Mean score | Approval rate | TPR | FPR | Accuracy |
|---|---|---|---|---|---|---|---|
| Under35 | 272 | 0.1360 | 0.1260 | 0.985 | 0.0000 | 0.01700 | 0.849 |
| 35to50 | 404 | 0.0866 | 0.0735 | 1.000 | 0.0000 | 0.00000 | 0.913 |
| Over50 | 324 | 0.0648 | 0.0647 | 0.997 | 0.0476 | 0.00000 | 0.938 |

*Table 2C. Outcomes by regional rating at threshold 0.50.*

| Group | n | Base default rate | Mean score | Approval rate | TPR | FPR | Accuracy |
|---|---|---|---|---|---|---|---|
| 1 | 101 | 0.0297 | 0.0474 | 1.000 | 0.0000 | 0.00000 | 0.970 |
| 2 | 743 | 0.0942 | 0.0818 | 0.996 | 0.0143 | 0.00297 | 0.904 |
| 3 | 156 | 0.1280 | 0.1250 | 0.987 | 0.0000 | 0.01470 | 0.859 |

A more informative perspective is to examine who bears the errors, revealing higher false positive rates for applicants under 35 and for regional group 3 (FPR = 0.0147), meaning they are more likely to be wrongly flagged as default and unjustly denied credit, which raises equal opportunity concerns because false positives correspond to unjust denials in an access to credit setting (Hardt et al., 2016; Barocas et

al., 2019). Figure 2 helps explain these patterns: men are shifted to higher predicted risk than women, younger applicants have higher risk scores than older applicants, and regional group 3 is shifted right relative to group 1. These patterns could reflect genuine risk heterogeneity, but they could also arise from label bias, measurement differences, or the use of proxy variables that indirectly encode sensitive characteristics.

From an ethical perspective, these false positives are not symmetric mistakes. A wrongful denial can create immediate hardship by delaying access to funds and can also carry reputational and psychological costs, especially for applicants who are early in their financial lives and have limited buffers. Because credit histories are path-dependent (label bias), being denied today can reduce a person's ability to build a record and can make future borrowing more expensive, so error rates that fall more heavily on younger applicants or certain regions can translate into cumulative disadvantage over time. This highlights that threshold choice is not neutral: a single global threshold can inadvertently concentrate harms on vulnerable groups, even when headline accuracy or apparent parity seems acceptable. Observed disparities should therefore be treated as prompts for investigation rather than as self-interpreting evidence of discrimination (Gebru et al., 2018).

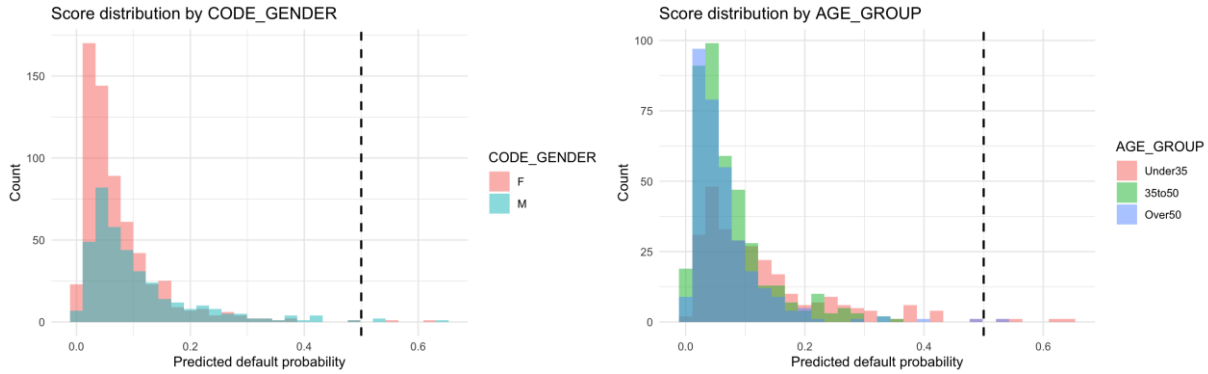



*Figure 2A. Predicted score distribution by gender.* *Figure 2B. Predicted score distribution by age band.*

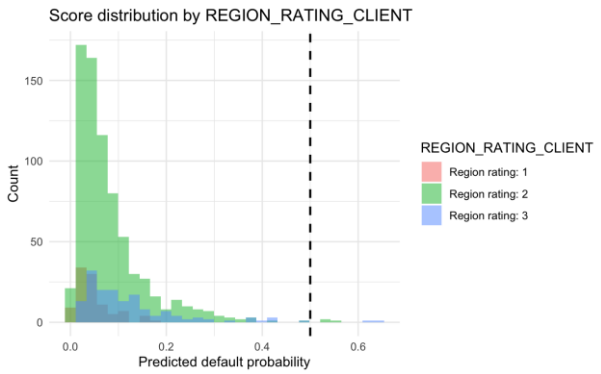



*Figure 2C. Predicted score distribution by regional rating.*

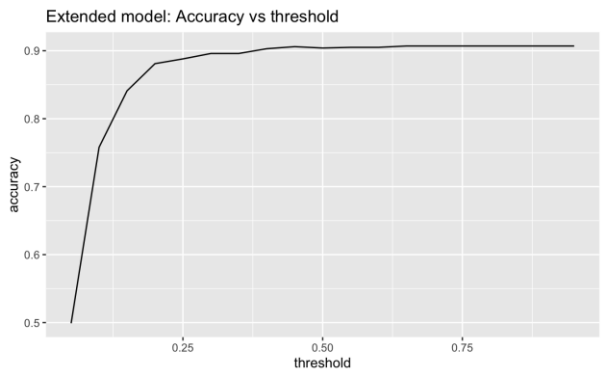
## 6. Threshold choice as an ethical lever

The threshold sweep illustrates the impact of decision design: low thresholds flag many applicants as default, approvals fall sharply, and gaps in selection rates and error rates can widen. When the threshold rises, accuracy increases and some parity gaps shrink, although part of this shrinkage is mechanical because the model predicts default less often. At thresholds at or above 0.50, the model flags very few applicants as default, so approval rate gaps become small even though default detection is weak.

When the threshold is set very low, denials rise, and the system becomes more exclusionary. In that regime, the primary ethical risk is that false rejections will be concentrated on applicants whose data profiles reflect structural disadvantage, so the model can operationalise a form of statistical redlining even without explicit protected attributes. Repeated rejection is not only a financial outcome but can also affect well-being through stress and loss of agency, and it can reinforce historical bias if the model is trained on outcomes shaped by earlier unequal access to credit.

When the threshold is set high, the model denies very few applications and parity in approval rates can look reassuring. However, the screening function is weakened, and risk is deferred rather than removed. Borrowers who face liquidity constraints or who rely on credit to smooth short-term shocks can be disproportionately exposed to loans that become unaffordable, so the long-run burden of default can still fall unevenly even when the short-run approval rates are similar. In this sense, threshold choice allocates power: it determines whether the institution prevents harm by restricting risky lending, or whether it permits risk to materialise later in ways that are harder for individuals to contest or avoid.

This report treats 0.50 as the status quo operating point, which can be defended as a presumption of creditworthiness. In effect, it is a default to yes rule. Unless the model assigns a probability of default above 0.50, the applicant is approved. That choice prioritises a low false positive rate for default prediction since false positives correspond to wrongful denials. It therefore values the applicant's right to access credit over the lender's ability to capture every true default. The ethical claim is not that this threshold maximises prediction, but that it encodes a conservative decision logic that avoids exclusion unless evidence of risk is strong (Hardt et al., 2016).



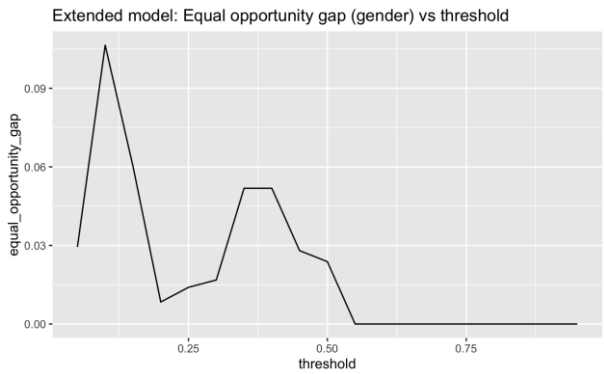




*Figure 3A. Overall accuracy versus threshold.* *Figure 3B. Equal opportunity gap (gender) versus threshold.*

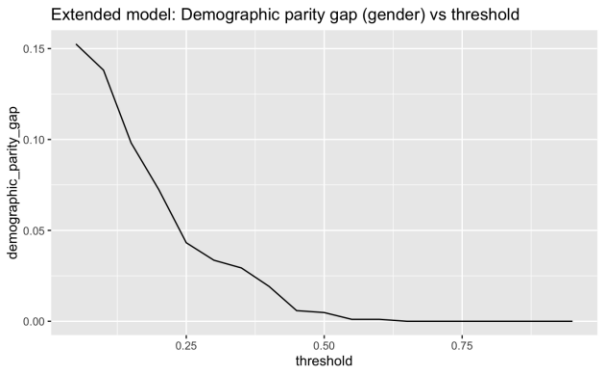



*Figure 3C. Demographic parity gap (predicted-default rate gap between genders) versus threshold.*

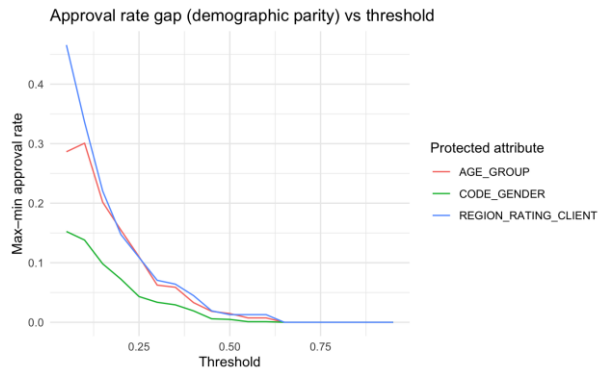
| threshold <dbl> | accuracy <dbl> | demographic_parity_gap <dbl> | equal_opportunity_gap <dbl> |
|---|---|---|---|
| 0.05 | 0.499 | 0.152533333 | 0.029411765 |
| 0.10 | 0.758 | 0.138133333 | 0.106442577 |
| 0.15 | 0.841 | 0.098133333 | 0.060224090 |
| 0.20 | 0.881 | 0.072533333 | 0.008403361 |
| 0.25 | 0.888 | 0.043200000 | 0.014005602 |
| 0.30 | 0.896 | 0.033600000 | 0.016806723 |
| 0.35 | 0.896 | 0.029333333 | 0.051820728 |
| 0.40 | 0.903 | 0.019200000 | 0.051820728 |
| 0.45 | 0.906 | 0.005866667 | 0.028011204 |
| 0.50 | 0.904 | 0.004800000 | 0.023809524 |

| demographic_parity_gap <dbl> | equal_opportunity_gap <dbl> | equalized_odds_FPR_gap <dbl> |
|---|---|---|
| 0.152533333 | 0.029411765 | 0.162287723 |
| 0.138133333 | 0.106442577 | 0.127643323 |
| 0.098133333 | 0.060224090 | 0.092292641 |
| 0.072533333 | 0.008403361 | 0.073746220 |
| 0.043200000 | 0.014005602 | 0.041675822 |
| 0.033600000 | 0.016806723 | 0.033331241 |
| 0.029333333 | 0.051820728 | 0.026064392 |
| 0.019200000 | 0.051820728 | 0.014533697 |
| 0.005866667 | 0.028011204 | 0.002521685 |
| 0.004800000 | 0.023809524 | 0.002521685 |

*Table 3. Representative thresholds from the sweep (gender).*

Ethically, this pattern shows why threshold choice is a value-laden decision. A system can appear fair on some metrics by switching off meaningful screening, yet that may simply shift harm elsewhere. If near-universal approval leads to defaults concentrated among already vulnerable borrowers, the model can still contribute to injustice even while headline parity measures look reassuring. For that reason, the threshold should be interpreted as a policy parameter that reflects social priorities, not as a purely statistical convention (Barocas et al., 2019).

Cost-sensitive learning offers a practical bridge between these views. It treats threshold selection as an optimisation of an explicit loss model rather than a convention such as 0.50. In credit settings, that objective can be framed in terms of the different harms associated with false approvals and false denials, including lender losses, consumer protection goals, and borrower harm from unaffordable debt (Elkan, 2001). However, this report does not implement a formal loss function or optimise thresholds under a cost model. Instead, the discussion is used to clarify that any fixed threshold implicitly encodes value judgements about which errors matter more, and fairness evaluation can be interpreted in that light rather than treated as a purely post hoc scorecard (Thomas et al., 2002).



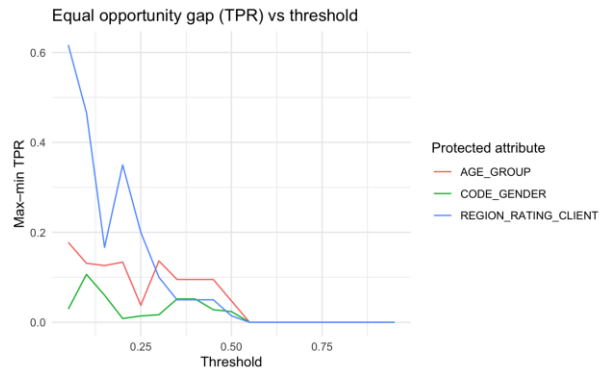




*Figure 4A. Threshold sweep fairness gaps by gender. Figure 4B. Threshold sweep fairness gaps by age band.*

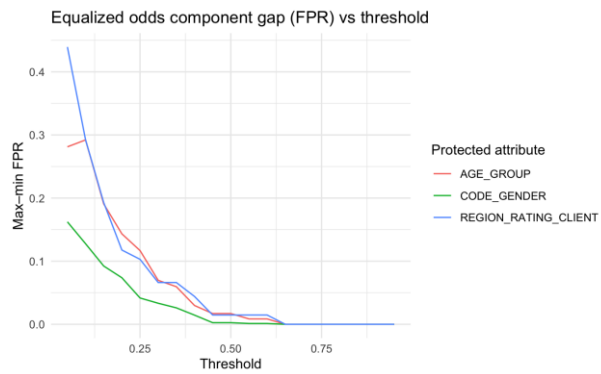



*Figure 4C. Threshold sweep fairness gaps by regional rating.*

## 7. Calibration and its distributive meaning

Calibration asks whether predicted probabilities correspond to observed default frequencies. It matters because credit decisions influence not only accept-reject outcomes, but also shape pricing, credit limits, or targeted support, giving probability quality direct distributional consequences. For that reason, calibration can be read as a form of distributive justice.

Figure 5 reports the overall calibration curve for the extended model using deciles of predicted risk. Points close to the 45-degree line indicate that predicted probabilities align with observed default rates within each decile, while deviations from the line indicate miscalibration. Points above the diagonal suggest under-prediction of default risk, and points below suggest over-prediction (Niculescu-Mizil and Caruana, 2005). When a group's risk is systematically under-predicted, members of that group may be offered credit on terms that do not reflect their repayment capacity. When risk is systematically over-predicted, the same group may face unnecessarily strict terms or exclusion. In this sense, calibration

connects to actuarial fairness, since it concerns whether risk is being measured consistently across groups (Brier, 1950).

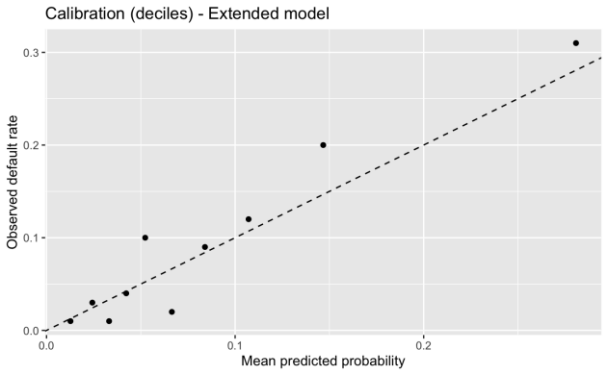



*Figure 5. Overall calibration plot (deciles) for the extended model.*

Figure 6 shows group calibration, highlighting systematic differences: women, men (the mean predicted risk is below the observed default rate), and applicants under 35 are generally under-predicted, while regional group 1 is over-predicted. If probabilities feed into pricing or limits, these patterns can create implicit cross subsidies where some groups are overcharged relative to realised risk, and others are undercharged. Even if the approval threshold is fixed, miscalibration can therefore generate fairness concerns, violating principles of distributive justice and accountability by assigning risk and costs unevenly across groups, because it shapes how risk is attributed and acted upon (Niculescu-Mizil and Caruana, 2005; Pleiss et al., 2017).

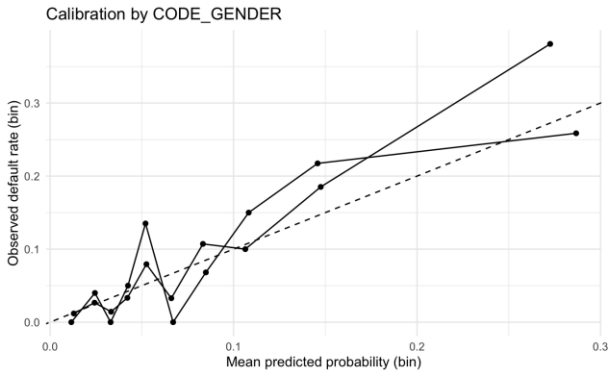



*Figure 6A. Calibration by gender.*

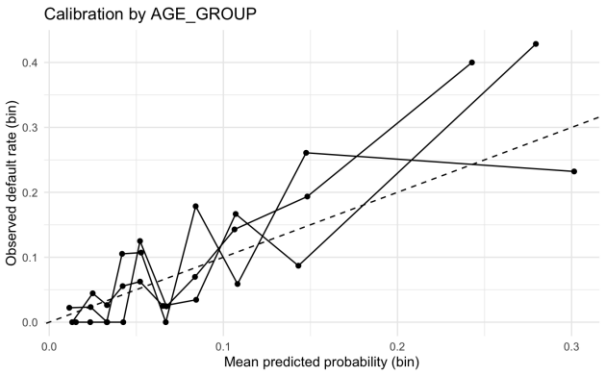



*Figure 6B. Calibration by age band.*

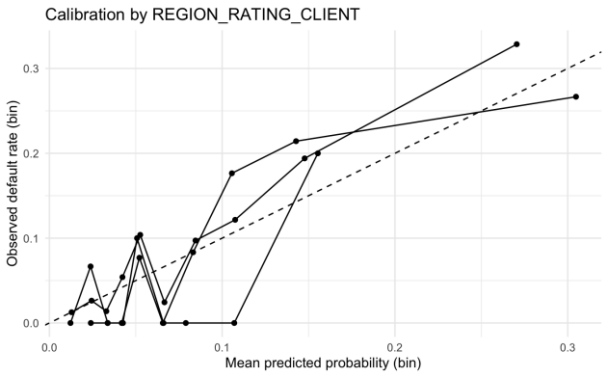



*Figure 6C. Calibration by regional rating.*

Calibration also interacts with fairness constraints. When base rates differ, one typically cannot achieve both within-group calibration and equalised odds under a single common threshold (Kleinberg et al., 2016; Pleiss et al., 2017). A similar tension appears for disparate impact style constraints (Chouldechova, 2017). This is another reason why fairness needs to be framed as a choice among imperfect options rather than as a search for a single best metric.

## 8. Ethical interpretation and governance implications

Taken together, the results support three connected claims. First, richer features improve predictive discrimination. In this analysis, the AUC improvement coincides with the inclusion of the external score variables (EXT_SOURCE_1–3), and in the fitted extended model, these variables show a strong statistical association with default risk. The reliance on external features does not inherently ensure fair outcomes. This underscores the tension between instrumental value, such as accuracy, and intrinsic values, including equity and transparency, which becomes particularly salient when improvements in prediction interact with social consequences such as access and fairness. Second, threshold choice dominates the distribution of harms and benefits. At 0.50, the model looks fair in approval rates mainly because it denies almost nobody, yet it fails to identify most true defaults. In practice, it deliberately trades off predictive precision against the social impact of automated denials, demonstrating how operational parameters encode normative judgments, and showing that these technical settings inherently involve ethical trade-offs. Third, even at this lenient operating point, disparities remain visible in score distributions, false positives, and calibration across age and region.

These findings matter because credit scoring is a socio-technical system. The model is one component, but the surrounding policy determines what counts as success, what is treated as an acceptable error, and how accountability is allocated. Documentation practices such as model cards and datasheets help keep these choices visible and contestable (Gebru et al., 2018; Mitchell et al., 2019).

The reliance on EXT_SOURCE variables strengthens the case for governance safeguards. If performance gains depend on external scores whose construction is not transparent, then the lender should require documentation about what data sources were used, how protected attributes were handled, and how errors are corrected. Model reporting tools and dataset documentation practices are designed to make these dependencies visible and, by doing so, to reduce the risk of erroneous decisions or unfair denials caused by model opacity, thereby mitigating potential harm to users or society, so they should be treated as part of deployment rather than as optional extras (Gebru et al., 2018; Mitchell et al., 2019).

A practical governance implication is that the model should not be the final decision-maker. The 0.50 threshold is intentionally non-exclusionary and can serve as a fast-track approval boundary, with low-risk applicants automatically approved and mid-risk cases routed to manual review, allowing loan officers to consider context, affordability, and dispute rights. High-risk cases trigger automated stops, limiting the scope of fully automated adverse decisions while maintaining accountability and oversight because it limits the scope of fully automated adverse decisions and creates a clear process for human oversight and explanation (ICO, 2023; Mitchell et al., 2019). This process minimises social harm by reducing the risk of unfair denials for moderate-risk applicants. Only applicants at or above 0.50 would face an automated stop under this design, which justifies keeping 0.50 in the code as a hard boundary for automation rather than as the main operational threshold.

Governance should also include ongoing monitoring for subgroup harms, particularly differences in false positives when denial is the main immediate harm (Hardt et al., 2016; Barocas et al., 2019). Where disparities are judged unacceptable, mitigation can occur at different stages. Pre-processing methods aim to reduce discrimination in the training data, for example, through reweighting (Kamiran and Calders, 2011; Dwork et al., 2011). In processing approaches, optimise prediction subject to fairness constraints (Agarwal et al., 2018). Post-processing methods adjust thresholds or recalibrate scores, although these interventions still encode normative trade-offs that must be defended rather than assumed (Pleiss et al., 2017).

Monitoring should also track the overall approval and rejection rates over time. If the organisation adjusts thresholds primarily to improve apparent accuracy or to meet a single parity target, it can unintentionally create either mass exclusion or overly permissive lending. A more defensible approach is to treat approval rates as part of consumer protection, set them in relation to an explicit cost model, and document the rationale so that decisions remain explainable and contestable.

Regulatory and due process considerations remain central. UK guidance emphasises accountability, lawful processing, and fairness in automated decision-making (ICO, 2023; Mitchell et al., 2019). In practice, this supports a cautious stance on sensitive and proxy variables unless there is a clear justification, a meaningful explanation to affected individuals, and an effective route for contestation.

## 9. Limitations and next steps

This analysis has limitations that affect ethical interpretation. The most important is class imbalance. Since defaults are rare, standard summaries such as accuracy can be misleading, and subgroup error rates can be unstable when the number of defaults in a group is small. The sample size of 5,000 supports a reproducible audit, but subgroup estimates can still be noisy, particularly for smaller regional groups. A further limitation is that default labels reflect realised outcomes under historical lending policies. If earlier policies were unequal, labels may encode past exclusion or differential exposure to risk. Finally, the chosen 0.50 threshold is ethically cautious for borrowers because it reduces wrongful denials, but it is financially risky for lenders and may also expose borrowers to harm if unaffordable loans are approved and later default (Barocas et al., 2019; King and Zeng, 2001).

Future work could strengthen the analysis in three ways. First, threshold choice could be tied to an explicit cost model that incorporates borrower harm and consumer protection objectives (Elkan, 2001; Thomas et al., 2002). Second, fairness audits could consider intersectional groups rather than one attribute at a time (Dwork et al., 2011). Third, a causal analysis could be added to test proxy mechanisms and to clarify which variables are ethically permissible risk factors in context (Kleinberg et al., 2016).

## 10. Conclusion

The extended logistic regression improves predictive discrimination, but the ethical evaluation depends on end-to-end decision design. At a 0.50 threshold, the system denies almost nobody, so approval rates

appear similar across groups. However, default detection is extremely weak, and disparities still appear in false positives and calibration across age and region. The broader conclusion is that responsible credit modelling requires joint attention to prediction, cost-sensitive thresholding, subgroup error profiles, and calibration. These technical choices should be paired with transparent documentation, monitoring, and mitigation options that are aligned with governance expectations and with a defensible ethical rationale.

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